

CogniHire: Auditing Résumé Claims Instead of Scoring Candidates

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Abstract—Automated hiring systems overwhelmingly emit a *score*: a scalar that ranks a candidate against others. Such scores resist explanation, give a candidate nothing specific to contest, and carry growing regulatory exposure under New York City Local Law 144, the Illinois Human Rights Act as amended by HB 3773, and the EU Artificial Intelligence Act. We present CogniHire, a technical screening system built on the opposite premise: it audits the claims a candidate makes and never scores the candidate. Three technology families carry the load. Evidence grounding constrains a language model to *select* claim text and forbids it to *author* any; across roughly 100,000 adversarial trials over 5,200 résumés the gate rejected every meaning-preserving paraphrase and every negation trap while admitting every verbatim claim. For machine learning we report a nine-model, six-family comparison under grouped cross-validation with significance testing, and a within-query diagnostic that separates matching ability from candidate-level quality, which pooled metrics cannot do. For deep learning we show that a plausible authored face-similarity threshold of 0.50 would have falsely rejected 41.4% of genuine identity matches, against 3.4% at a threshold calibrated on Labeled Faces in the Wild. The recurring engineering move is converting policy into structure: an import boundary that fails the build, a type with no field to hold a fabricated value, a vocabulary ban that forbids a composite score.

Index Terms—automated hiring, evidence grounding, claim verification, algorithmic accountability, model comparison, face verification, threshold calibration

I. INTRODUCTION

A technical interview ends and a form is filled in: “*Strong communication. Good Flutter knowledge. 8.7/10.*” Six months later the number is indefensible. Nobody can reconstruct which answer moved it from 8.4 to 8.7; the candidate cannot appeal it and the recruiter cannot defend it. The number was a compression of knowledge that discarded everything needed to audit it.

Automated interview platforms largely reproduce this failure at scale. Faced with criticism, the industry has made the scalar more sophisticated—more features, better models, fairness post-processing—but few have asked whether a scalar was the right output object at all.

We take the opposite position: the useful output of automated screening is an audit of the claims a candidate has made, not a judgment of the candidate. A résumé is a set of assertions. Each can be probed, each probe yields evidence, and evidence supports a verdict about *the claim*. The verdicts,

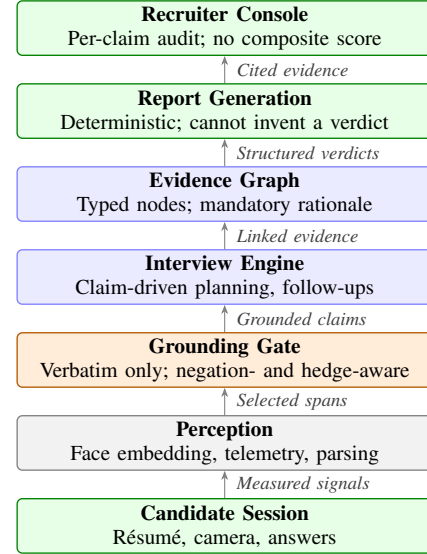


Fig. 1. Data flow in CogniHire. Evidence moves upward from the candidate session to the recruiter console; each transition narrows what the system may assert. The guarantees enforced at these boundaries are listed in Table I.

with evidence attached, go to a human who decides about *the person* (Fig. 1). This changes what the system must be *unable* to do, and which technology belongs at which layer: grounding governs what may be asserted, machine learning (ML) measures properties of evidence, and deep learning (DL) supplies perception.

Our contributions are:

- 1) A **claim-audit output model** replacing the composite score, with a four-valued verdict set that makes unmeasured evidence explicit.
- 2) A **structurally enforced grounding constraint**, measured at 100% paraphrase and negation-trap rejection over $\sim 100,000$ trials.
- 3) A **nine-model comparison** under grouped cross-validation with Friedman and Holm-corrected Wilcoxon tests, plus a synthetic positive control establishing statistical power.
- 4) A **within-query ranking diagnostic** that separates matching from entity-level quality on any paired benchmark.

TABLE I
POLICIES CONVERTED INTO STRUCTURAL GUARANTEES

Policy	Structure
AI must not author claims	Import boundary plus a CI architecture test
Unmeasured is not passed	Result type with no similarity field
No composite score	Vocabulary ban that fails the build
No demographic inference	Classifier module never loaded (allow-list)
No outcome labels	No such dataset, no code path to build one
Plans cannot be edited later	Question plans never persisted
Sessions reconstructible	Append-only event table with sequencing trigger

- 5) An **empirical threshold calibration** showing that a documented, authored constant was wrong by a quantified amount.

II. RELATED WORK

Raghavan et al. [1] surveyed vendors of algorithmic pre-employment assessment and highlighted the consequences of a vendor’s choice of prediction target: fairness analysis is conducted against the stated target rather than the deployed one. They offer no remedy for a misdescribed target; Section V-B supplies a measurement that detects one.

A second strand re-ran published comparisons under controlled protocols. Ferrari Dacrema et al. [2] could reproduce only seven of eighteen neural recommenders, six of which were often beaten by simple heuristics; Musgrave et al. [3] found the same for deep metric learning, and Grinsztajn et al. [4] showed tree ensembles still rival deep networks on tabular data. We differ in finding that no method separates from any other, and then tracing why.

On evaluation protocol, Demšar [5] cautions against reading a ranking off raw score differences; Nadeau and Bengio [6] show that overlapping training sets bias naive variance estimates; Gorman and Bedrick [7] showed rankings from a single standard split often fail to reproduce; and Kapoor and Narayanan [8] documented leakage in 294 ML-based science papers. We adopt Demšar’s protocol and quantify one leakage category on a hiring benchmark.

For perception we build on SCRFD [9] and ArcFace [10], which leave the verification threshold to the integrator, and calibrate it on LFW [11]. Across these strands three gaps recur: no work constrains generative extraction to selection; prediction targets go unverified behind pooled metrics; and perception components ship with authored, unmeasured thresholds.

III. SYSTEM ARCHITECTURE

CogniHire spans four surfaces: a candidate portal (consent, résumé, session media), an inference service (perception and learned components), an event-sourced persistence layer, and a recruiter console that renders the audit. Every guarantee is enforced at a boundary between two surfaces. Session state is written to an append-only table with a sequence-assigning trigger, so a session can be reconstructed turn by turn; question

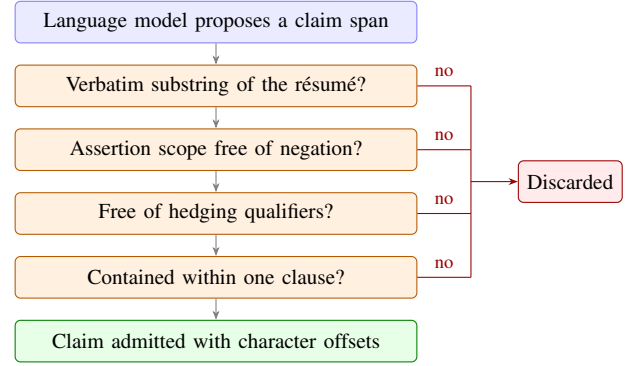


Fig. 2. The grounding gate as a decision pipeline. Any *no* branch discards the proposed claim.

plans are never persisted, so a plan cannot be edited after the fact to match an outcome.

The audit has four verdicts:

- substantiated,
- notDemonstrated,
- contradicted, and
- notExamined.

The four are deliberately not orderable, and no composite is computed. *notExamined* is load-bearing: without it an unprobed claim is simply absent from the report, and absence reads as acceptance.

Table I lists the policies the system converts into structures. Each turns a claim that requires trust into one a skeptical reviewer can check by inspection.

IV. EVIDENCE GROUNDING

A. The Grounding Gate

The central rule: a claim’s text must be a verbatim substring of the résumé. The model selects; it does not generate. Naive substring matching is insufficient, so the gate (Fig. 2) adds negation rejection (“I have not worked with Kubernetes” yields no Kubernetes claim), hedge rejection (“some exposure to Kafka”), contrastive-conjunction splitting (“I used Django, but never in production”), and clause scoping, so matching never crosses a clause boundary.

The grounding module is forbidden from importing the AI module, and an architecture test fails continuous integration if that import appears. The guarantee is a property of the build, not of developer discipline. Because the planner consumes only typed claims, an instruction embedded in a résumé has no field to occupy, which also closes the prompt-injection channel.

B. Measured Behavior

An offline deterministic harness drives the real pipeline modules over 5,200 synthetic résumés spanning eight fields and four seniority levels. No language-model key is needed, so the result is reproducible. Across roughly 100,000 trials the gate rejected all 48,149 meaning-preserving paraphrases and all 51,761 negation traps (Table II).

TABLE II
GROUNDING-GATE CORRECT-DECISION RATE BEFORE AND AFTER THE
CLAUSE-SPLITTER FIX

Trial type	n	Before fix	After fix
Paraphrases (must reject)	48,149	100%	100%
Negation traps (must reject)	51,761	100%	100%
Verbatim claims (must admit)	51,761	99.18%	100%

TABLE III
NINE CLASSIFIERS, TEN-FOLD CROSS-VALIDATION GROUPED BY
POSTING (MEAN $\pm$ FOLD STD.)

Model	AUROC	ECE	Lat. (μ s)	Size (KB)
MLP	0.696 $\pm$ 0.031	0.057	0.5	74.7
Gradient boosting	0.689 $\pm$ 0.028	0.047	1.9	393.3
Random forest	0.688 $\pm$ 0.030	0.047	88.2	24,732.6
SVM (RBF)	0.686 $\pm$ 0.028	0.053	195.0	526.8
Extra trees	0.685 $\pm$ 0.030	0.046	84.9	63,415.9
k -NN	0.680 $\pm$ 0.031	0.057	35.5	1,272.8
Hist. grad. boosting	0.679 $\pm$ 0.029	0.057	7.3	591.8
Gaussian NB	0.667 $\pm$ 0.033	0.117	0.3	1.1
Logistic regression	0.665 $\pm$ 0.022	0.052	0.2	0.9

The initial run showed a 0.82% false-rejection rate: 425 of 51,761 genuine claims discarded, every one containing a period-bearing token such as `Node.js`, `asp.net` or `Python 3.9`. The clause splitter treated every period as a terminal, so “`Node.js`” split into “`Node`” and “`js`.” and the claim was *silently* dropped. Requiring a terminal to be followed by whitespace or end-of-string fixed it; since the change only makes clauses longer, it widens negation scope and cannot weaken the guard. Verbatim admission rose to 100% with both rejection rates unchanged. No unit test caught this defect; only corpus-scale measurement with an explicit true-positive metric did.

The corpus is synthetic and single-generator, so we do not claim 100% extraction coverage in general. The defensible claim concerns the safety mechanism: the model selects and never authors.

V. MACHINE LEARNING: WHAT MODEL CHOICE BUYS

ML in CogniHire measures properties of evidence; it is never trained on hiring outcomes. To test how much algorithm choice matters in screening, we compared nine classifiers from six families on a public résumé-posting fit benchmark [12] of 7,910 balanced pairs (643 résumés, 351 postings), using scikit-learn [13]. The dataset card states no description, license, or labeling procedure, which bounds how far any result on it generalizes.

A. Protocol and Results

Every experiment uses ten-fold `GroupKFold` grouped by posting. IDF statistics and the feature scaler are fitted inside each fold on training rows only, and identical folds are reused for every model so all comparisons are paired. Table III reports the results.

The nine span only 0.031 AUROC, comparable to each model’s fold-to-fold standard deviation. The Friedman test

TABLE IV
PAIRWISE RANKING ACCURACY (RANDOM FOREST, 4,112
WITHIN-RÉSUMÉ PAIRS, CHANCE = 0.5)

Representation	Between	Within	Binomial p
Relational (10-d)	0.6905	0.6333	3.8×10^{-66}
Interaction (1536-d)	0.7372	0.6240	2.0×10^{-57}

rejects identical performance ($\chi^2 = 38.35$, $p = 6.5 \times 10^{-6}$), but under paired Wilcoxon tests with Holm correction the leading MLP is **statistically indistinguishable from all four tree ensembles**. A ranked table alone would have licensed “MLP is best”; the significance analysis does not.

Power. A null is uninformative unless the protocol can detect a difference when one exists. On a synthetic control with a planted generative process (Bayes-optimal accuracy 0.925), the same nine models under the identical code path separate decisively ($\chi^2 = 55.42$, $p = 3.7 \times 10^{-9}$; Fig. 3). The null is a property of the task, not the protocol.

Tuning. Nested cross-validation with an equal eight-configuration budget per model improves no model by more than 0.0077 AUROC, and after tuning the leader separates from fewer rivals, not more.

B. Separating Matching From Candidate Quality

A pooled metric rewards two capabilities it cannot distinguish: knowing which résumés tend to be labeled fit, and knowing which postings a given résumé fits. Only the second is matching. We varied the input representation under identical folds. The best representation, concatenated 1536-dimensional résumé and posting embeddings, reaches 0.755 AUROC. A probe given only the résumé embedding—which cannot be matching—reaches 0.660 AUROC, 87% of that best score.

This suggested the labels were not relational at all. A direct test refuted that: 80.6% of label variance is within-résumé. We therefore evaluate on pairs of rows sharing a résumé, one positive and one negative, and ask whether the model ranks the positive posting higher. Every résumé-level property cancels, so chance is exactly 0.5.

Models do match: given one résumé and two postings, they pick the fitting one about 63% of the time (Table IV). But for the same random forest, moving from the relational to the 1536-dimensional interaction representation raises pooled AUROC by +0.048 (0.688 to 0.737) and between-résumé ranking by +0.047, while within-résumé matching changes by -0.009 ($p > 0.4$, paired Wilcoxon over folds). The improvement a leaderboard would record came from ranking candidates against each other, not from matching them to roles. The diagnostic needs no extra labels and applies to any paired benchmark with a recurring entity (patient–treatment, user–item, query–document). In hiring, it is precisely the measurement that distinguishes ranking matches from ranking people—the target substitution Raghavan et al. [1] identify.

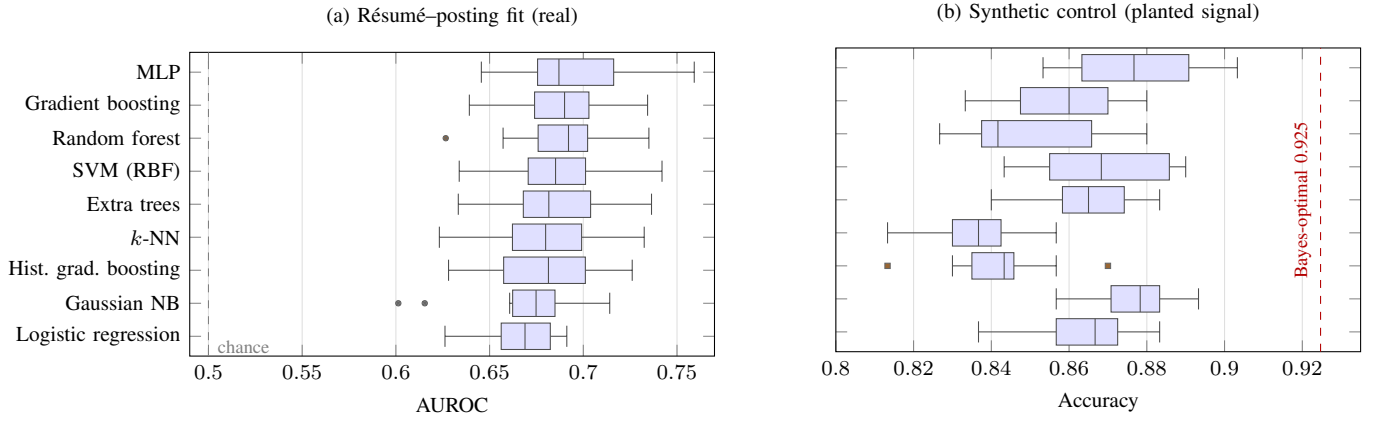


Fig. 3. Per-fold results over ten grouped folds. (a) Nine models on the real screening task (AUROC): boxes overlap almost completely. (b) Same models, folds, and code on the synthetic positive control (accuracy), which they separate clearly.

TABLE V
SPLIT-SCHEME COMPARISON ON ALL 8,000 ROWS (CLASS-WEIGHTED
RANDOM FOREST, RELATIONAL FEATURES; MEAN $\pm$ FOLD STD.)

Scheme	Held out	AUROC
k -fold, no grouping	nothing	0.714 ± 0.016
GroupKFold by résumé	every test résumé	0.703 ± 0.031
GroupKFold by posting	every test posting	0.689 ± 0.016

C. Robustness and Efficiency

Table V varies only the splitting scheme, holding features and model fixed. Because the same run also tests label granularity, it uses all 8,000 rows without the class-balancing downsample and a class-weighted 300-tree random forest on the relational features; its numbers are comparable with each other, not with Table III. The ungrouped scheme a conventional bake-off would use scores 0.024 AUROC above the posting-grouped protocol, a modest leakage inflation. Closing the résumé channel, however, does not collapse performance: grouping by résumé scores 0.014 *higher*, which refutes a strong-memorization account. The ordering tracks which side of the pair is novel, and generalizing to a new posting is the harder problem—exactly the condition every new requisition presents. Binarization is not hiding signal either: the cleanest contrast, Good Fit against No Fit with the ambiguous class removed, scores 0.681, no better than the 0.689 of the binarized label.

A learning curve over the same folds rises from 0.651 AUROC at a tenth of the data to 0.688 at all of it; doubling from half to full buys only 0.008 (Fig. 4). The curve flattens well below any useful operating point, so the limitation is not sample size.

Meanwhile inference latency spans three orders of magnitude (0.2 to 195.0 μ s per row) and serialized model size spans 0.9 KB to 63.4 MB (Table III), for 0.031 AUROC (Fig. 5); the rational choice is the cheapest model not significantly worse.

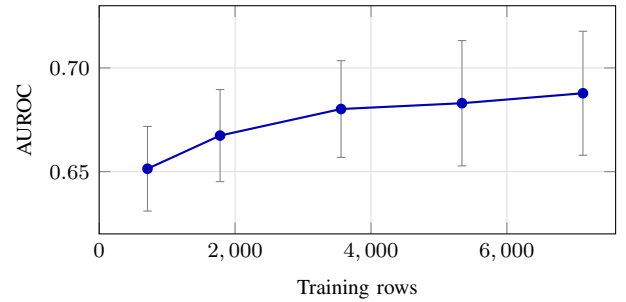


Fig. 4. Learning curve over the same ten folds (mean $\pm$ fold std.).

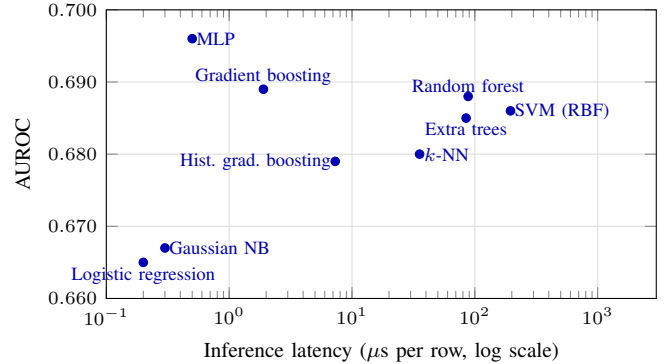


Fig. 5. AUROC against inference latency (log scale). Three orders of magnitude of latency buy 0.031 AUROC.

VI. DEEP LEARNING: CALIBRATING IDENTITY

Face detection uses SCRFD [9] and embedding uses ArcFace [10]; both are frozen. Retraining on project-scale data would be worse and would introduce demographic bias we cannot audit. The vendor pack’s age and gender classifier is excluded by a loader allow-list.

The system originally used an authored cosine-similarity threshold of 0.50, reasoned and documented but flagged as uncalibrated. An equal-error-rate sweep on 2,200 LFW [11]

TABLE VI
FACE-VERIFICATION THRESHOLD BEFORE AND AFTER CALIBRATION
(LFW, 1,000 HELD-OUT PAIRS)

Threshold	FAR	FRR	Acc.
0.1266 (calibrated)	0.030	0.034	0.968
0.50 (authored)	0.000	0.414	0.793

training pairs, evaluated on 1,000 held-out pairs, gives Table VI; the embedding’s AUC on the held-out pairs is 0.9785 at any threshold.

The authored value would have falsely rejected 41.4% of genuine matches. Because the system correctly refuses to treat an unmeasured check as a pass, roughly two in five legitimate candidates would have accumulated integrity flags for being themselves. Calibration cut false rejection twelvefold at a false-acceptance cost of 3.0%. In security-adjacent systems, *reasoned* and *correct* are different properties, and only one can be demonstrated.

The identity result type is three-valued. *Unchecked* carries a reason and has no similarity field at all—not 0.0, not null—so no downstream comparison can read a failed measurement as a low-but-real one, in the spirit of distribution-free abstention [14]. LFW is a public-figure benchmark, not our candidate population; population-specific recalibration requires consented candidate data we have not collected.

VII. INTEGRATION, COMPLIANCE, AND LIMITATIONS

Each family alone leaves a gap. Grounding guarantees claims are the candidate’s own words but not that a session gathered enough evidence to judge them; ML measures evidence sufficiency but, without grounding, would measure text a model may have invented; DL supplies identity continuity but no verdict. Together they close a loop that terminates in a human reviewer, not an aggregate.

This design maps directly onto regulation (Table VII). The instruments share a direction: an automated employment decision must be disclosed, auditable, and open to human oversight and explanation. A system architected around a scalar faces a retrofit its own architecture resists, since the explanatory information was discarded at design time. The mapping in Table VII is possible only because CogniHire declines to produce an aggregate: a bias audit of a composite score must explain the score, whereas a bias audit of a claim audit examines the distribution of verdicts and the evidence attached to each—a smaller and better-posed question.

Lessons from a reference implementation. Before building, we audited a comparable open system in full. Its computer-vision core was sound, but its decision layer was largely fabricated. An object detector configured with an invalid model name never ran once; a luminance heuristic filled the gap and logged window glare as *camera-verified* phone detections, which fed an escalating penalty system, while a status string reading *engine ready* was never rendered anywhere. The identity endpoint returned a 0.98-confidence match when no enrolled face existed at all, and a résumé analyzer returned

a hardcoded skill list on zero keyword matches. The defects share one root cause: a fallback that fabricates a measurement is worse than an outage. CogniHire’s responses—a value-free type for failed measurements, component health surfaced to a human, every output traceable to a source span—follow directly. For the same reason we deliberately do *not* build affect, stress, or deception estimation: none has defensible ground truth on a diverse candidate pool, and a stress percentage that silently changes which questions a candidate is asked is itself a hidden score.

Limitations. We state three gaps openly. Identity matching is implemented and tested but is currently only a presence gate on the production path, so continuous identity verification is not yet deployed. The face threshold is calibrated on a public benchmark, not candidates. And no structured recruiter validation study has been conducted, so the premise that reviewers prefer a claim audit to a score is untested; a controlled comparison measuring inter-reviewer agreement is the most valuable next study. Finally, a résumé-posting fit model from an earlier iteration of this work (0.657 AUROC) is deliberately not deployed: Section V-B shows that pooled fit scores on this benchmark largely reward ranking candidates rather than matching them.

VIII. CONCLUSION AND FUTURE WORK

CogniHire replaces the composite hiring score with a per-claim audit whose guarantees are enforced by architecture rather than by prompt discipline. Three findings stand out. The grounding gate rejected every meaning-preserving paraphrase and negation trap across $\sim 100,000$ trials while admitting every verbatim claim. A carefully reasoned similarity constant would have falsely rejected 41.4% of genuine identity matches; measurement reduced that to 3.4%. And nine classifiers proved statistically indistinguishable at the top of a public screening benchmark, while a within-query diagnostic showed that the pooled metric’s apparent gains came from ranking candidates rather than matching them. That last finding overturned our own initial hypothesis; we report the sequence because a diagnostic proves its power only when it refutes something.

Three steps follow directly from the limitations in Section VII: a controlled recruiter study comparing inter-reviewer agreement on a claim audit against a composite score; recalibration of the face threshold on consented candidate data; and continuous identity verification on the production path.

DATA AVAILABILITY

Analysis code, per-fold results, and figure sources are available from the author on request. The face study uses the public LFW benchmark and the grounding evaluation a synthetic corpus; no human subjects were involved.

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TABLE VII
MAPPING REGULATORY OBLIGATIONS ONTO ARCHITECTURAL MECHANISMS

Obligation	Instrument	Mechanism in CogniHire
Bias audit of the tool	NYC Local Law 144 [15]	No composite score to audit; the audited object is the per-claim verdict distribution
Candidate notification	NYC Local Law 144 [15]; Illinois AI Video Interview Act [16]; Illinois HB 3773 [17]	Consent captured per session with a recorded consent version
Human oversight	EU AI Act, Art. 14 (high-risk systems; employment is listed in Annex III) [18]	Report generation is a deterministic transform containing no code that could emit a recommendation
Explanation of decisions	EU AI Act, Art. 86 [18]	Every verdict carries its evidence and character offsets into the candidate's own words
Non-discrimination	Illinois Human Rights Act as amended by HB 3773 [17]	No demographic inference, no outcome labels, no historical decision pattern to inherit
Record-keeping	EU AI Act, Art. 12 [18]	Append-only event store; sessions reconstructible turn by turn

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